



Medreach

RIGHT HOSPITAL · RIGHT NOW

The right hospital, right now, **confirmed before you leave home**

AI-powered emergency coordination for Madhya Pradesh: understand the emergency, find a hospital that can treat it now, get it to accept, and guide the patient there.

MPOnline Idea & Innovation Hackathon 2026

Challenge 5 · Healthcare

Live demo: medreach-2.onrender.com · Code: github.com/prakharpurwal518-collab/Medreach



Introduction: what Medreach is

Medreach is an AI-powered emergency coordination platform for Madhya Pradesh. It understands an emergency described in Hindi, English or Hinglish, finds a hospital that can treat it right now using staff-verified data, gets that hospital to accept the patient, and guides the patient there with live tracking.

Challenge 5 • AI Innovation for Public Services & Citizen-Centric Governance • Domain: Healthcare



Citizens & families

Voice-first emergency help, no login needed, “I’m alone” mode



Hospitals

Verified status board, referrals with an AI pre-arrival note



Government of MP

Live, audited view of emergency capacity across hospitals

7 / 7

suggested technologies used

2

private portals: citizen & hospital

16

Bhopal-region hospitals in the demo

51

automated tests passing



Live and deployed: medreach-2.onrender.com

Scenario: chest pain at 11 pm in Bhopal

A 62-year-old man has chest pain and is sweating. Here is what happens today, and what happens with Medreach.



Today

1

Family drives to the nearest hospital

Chosen by distance, with no idea of who is on duty

2

No cardiologist on duty, ICU full

Discovered only at the gate

3

Referred onward: second trip

Another drive across the city at night

4

ER unprepared; golden hour lost

No notes, no bay, no specialist called in advance



With Medreach

1

Son speaks the emergency in Hindi

“Papa ko seene mein dard hai, pasina aa raha hai”

2

AI: may indicate heart attack

Needs 24×7 emergency + cardiologist + ICU

3

Bansal Hospital accepts, bay Resus-02

Named officer accepts before the family leaves

4

Ambulance tracked live

ER read the AI pre-arrival note in advance

Result: one journey instead of two, and the hospital is ready when the patient arrives. (Illustrative scenario; time saved will be measured in the pilot.)

Overview: why patients lose the golden hour



1,68,491

road-accident deaths in India, 2022

Source: MoRTH, "Road Accidents in India 2022". Heart attacks, strokes, burns and obstetric emergencies add many more.



Nearest ≠ capable

Maps don't know if the specialist is on duty, the ICU is free or the CT works now.



No trusted live status

Availability is unpublished or outdated, with no owner and no time stamp.



No acceptance first

Families arrive unannounced; refusals happen at the gate.



Panic & language

Hard to type forms in panic; many prefer to speak in Hindi.



People living alone

Cannot call hospitals, arrange transport and inform family at once.



Lost information

Allergies and ABHA / Ayushman details don't reach the ER doctor.

Who is affected



Patients & families

Wasted trips, refusals, delayed care



Rural citizens & ASHA workers

Long distances make a wrong choice costly



Hospital ER teams

Unannounced critical arrivals, crowding



Health department

No live picture of emergency capacity

Solution: one flow from emergency to confirmed bed

An emergency coordination platform, not a diagnosis app. Every step of our flowchart is implemented and live.



Hindi · English · Hinglish

Voice-first and built for panic; quick-pick chips and one question at a time

Only capable hospitals

Deterministic, explainable matching on staff-verified, time-stamped data

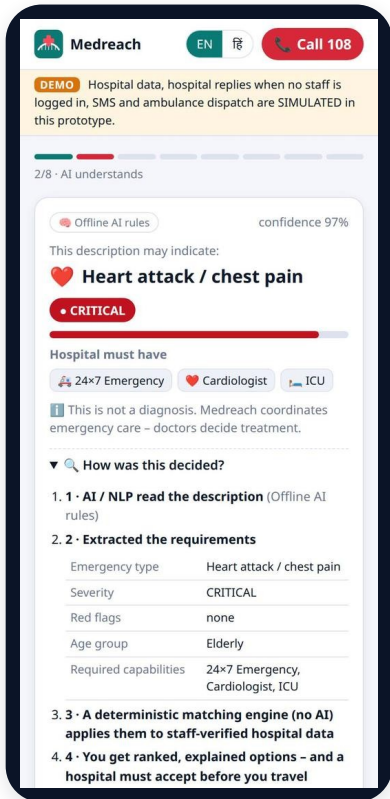
Accepted before travel

A named officer accepts and prepares a bay; no refusals at the gate

60 s auto-fallback

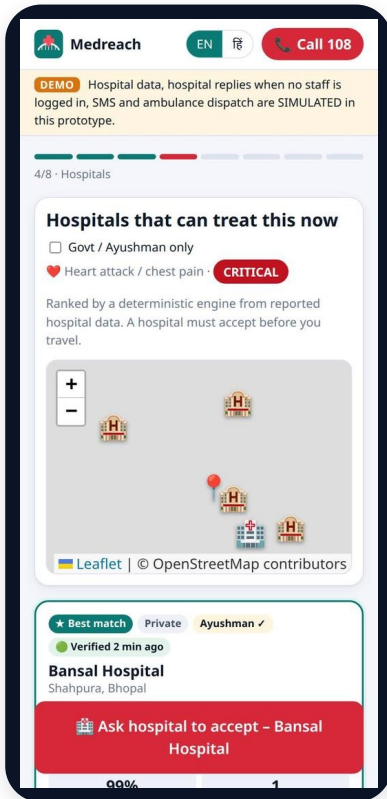
No answer or a decline → the next suitable hospital is asked automatically

Solution in action: citizen and hospital



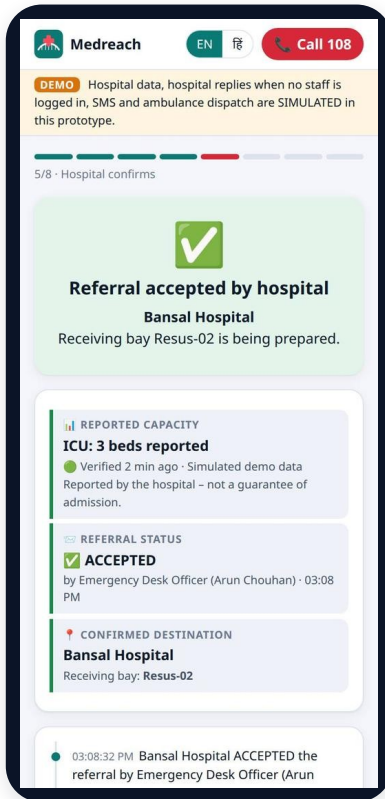
AI understands

May indicate heart attack · CRITICAL; “How was this decided?” shown



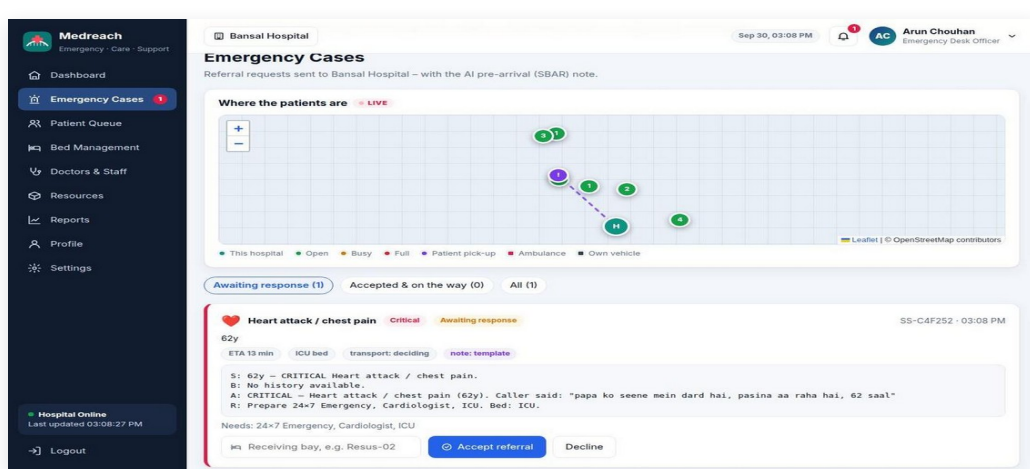
Capable hospitals

Ranked from verified data, with freshness badges



Accepted

Reported capacity → accepted → confirmed bay

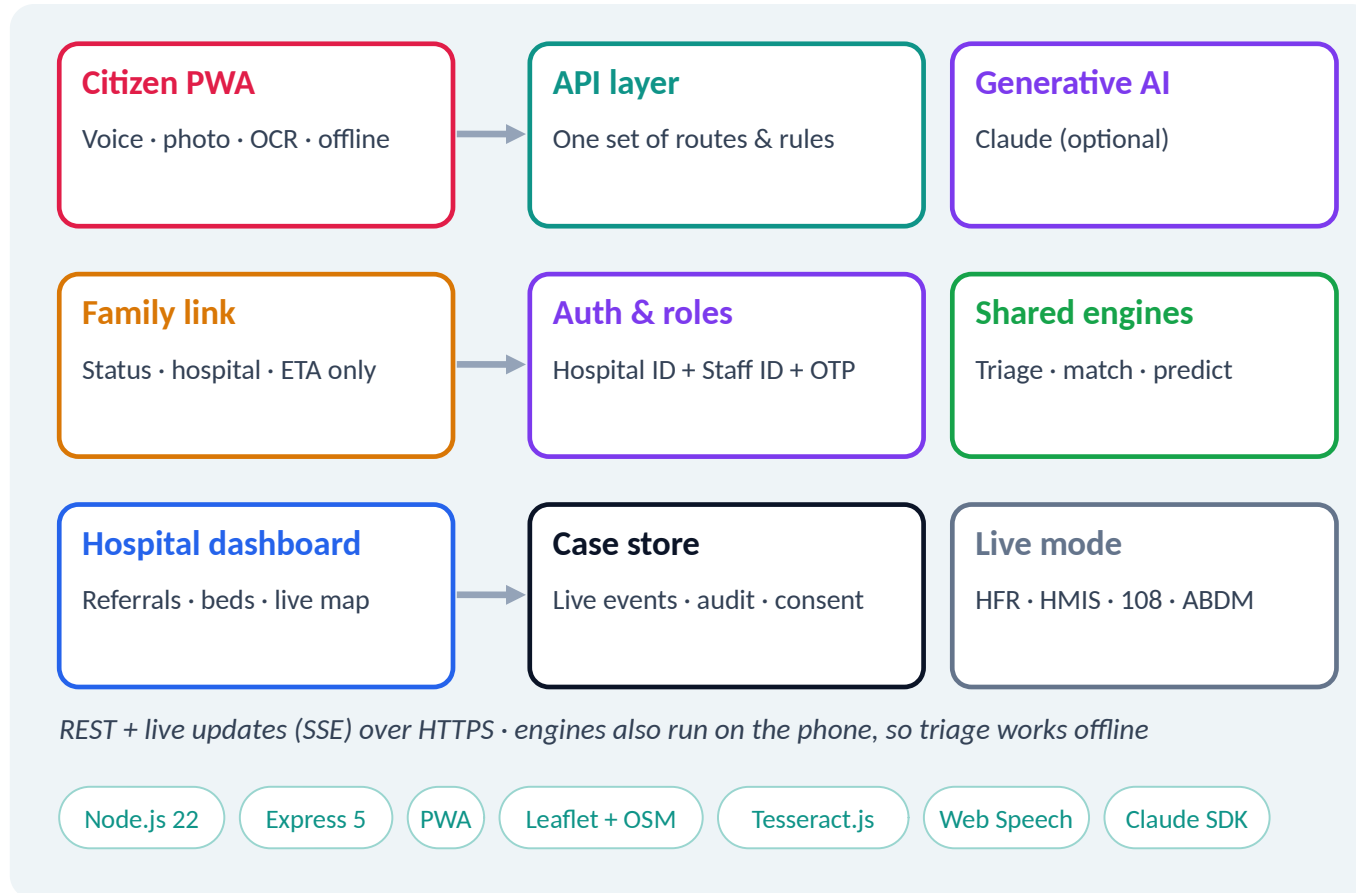


Hospital reviews the AI pre-arrival note and accepts with a receiving bay



Citizen dashboard: live map, ambulance on the way to the accepted hospital

Technical approach: architecture and AI



All 7 technologies in code

-  **Generative AI** requirements + SBAR note
-  **Voice Bots** Hindi / English in and out
-  **Computer Vision** photo of injury → findings
-  **OCR** Ayushman / ABHA card on phone
-  **Predictive Analytics** ETA, bed-on-arrival, forecast
-  **Conversational AI** one follow-up at a time
-  **Mobile App** offline, bilingual PWA

AI understands; rules decide. Severity = max(AI, rules), so AI can never lower urgency. A deterministic, tested engine chooses the hospital.

Security, privacy and trust by design

Who can change what citizens see

Role	Can do
Hospital Nodal Officer	Everything for their hospital + audit log
Emergency Desk Officer	Accept / decline referrals, ER open / busy / diverting
Resource Manager	Beds, specialists on duty, equipment
State Health Admin	Read-only view of all hospitals + audit logs



Hospital ID + Staff ID + OTP; session bound to one hospital



Staff accounts cannot be self-registered



Every login, change, decision and denied attempt is logged



Consent first

Patient details reach the hospital only if the family agrees.



Masked IDs

ABHA / Ayushman numbers masked; pick-up point rounded ~100 m.



Hospital-scoped access

Staff see only referrals sent to their own hospital.



Citizen data on device

Profile and records stay on the phone (PBKDF2 password hash).



Family link without medical data

Status, hospital and ETA only, via a read-only token.

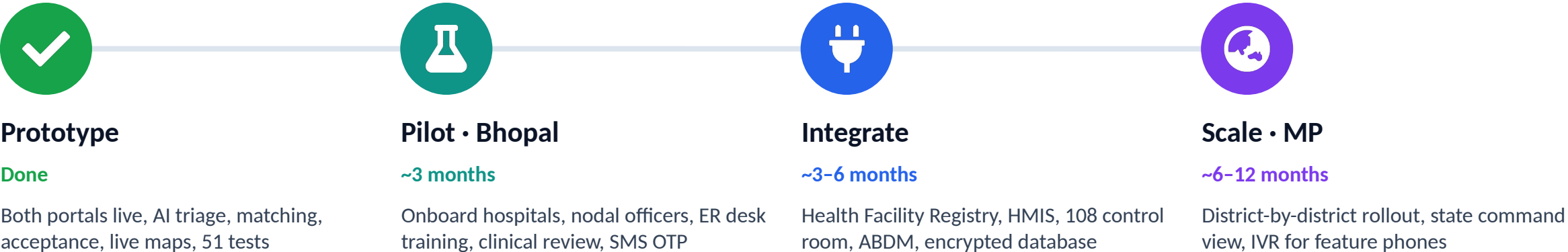


24-hour deletion

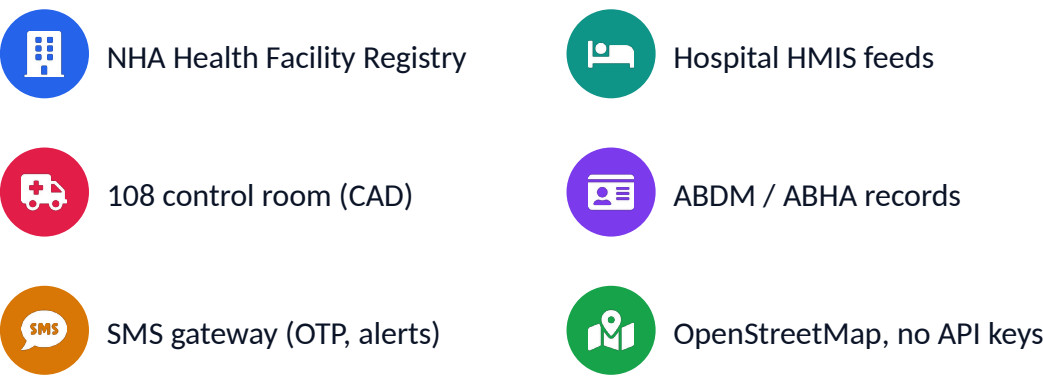
Personal and health details purged after hand-over.

Honest by design: reported capacity ≠ referral accepted ≠ confirmed destination; simulated demo data is always labelled. Aligned with the DPDP Act 2023.

Implementation: from prototype to statewide



Integrations for live mode



Risk	Mitigation
Hospitals don't update data	Freshness on every figure, stale ranked lower, one-click confirm
Wrong AI reading	AI only raises urgency; deterministic matching; 108 always visible
No hospital answers	60 s timeout → next suitable hospital
Weak network	Offline PWA runs triage & matching on the phone

Low cost

Fits existing roles

Cloud-ready

Innovation: what makes Medreach different

Capability	Map search	Bed portals	Helplines	Medreach
Understands the emergency (voice, Hindi)	No	No	Operator	AI + voice
Matches capability (specialist, equipment)	No	Beds only	Manual	Yes
Shows freshness & who updated data	No	Rarely	No	Every figure
Hospital accepts before travel	No	No	Sometimes	Yes + bay
Automatic fallback to next hospital	No	No	Manual	60 s
AI pre-arrival note for ER doctor	No	No	No	SBAR note
Help for a person living alone	No	No	No	"I'm alone"

Live proof: when a hospital sets its ER to "Diverting", it disappears from every citizen's list within a second.

< 1 s

hospital status change reaches every citizen

60 s

no answer → next hospital asked

0

API keys needed for the maps



Three truths kept separate

Reported capacity ≠ referral accepted ≠ confirmed destination.



Explainable score

"Why this hospital?" checklist first; every score part shown.



Privacy by design

Consent, masked IDs, audit log, deletion 24 h after hand-over.



Offline-first

Same engines run on server and phone; works on weak networks.

Impact and benefits



Social

Right care the first time; equal access for rural, elderly and Hindi-speaking citizens



Economic

No repeat trips or wasted visits; better use of ICU beds and specialists



Governance

Live, audited view of emergency capacity for planning beds and ambulances



Educational

Safe first-aid steps, "not a diagnosis" clarity, awareness of Ayushman, ABHA, 108



User

One button, voice, bilingual, clear status and a family tracking link



Health system

Fewer surprise arrivals, less ER crowding, fewer inter-hospital transfers

Benefits by stakeholder



Patient

Treated sooner, bay ready



Family

Live link, no medical data



ER team

Advance notice + AI note



Government

Audited capacity data



108 service

Destination already accepted



ASHA workers

Raise an emergency by voice

How we will measure it in the pilot



Time to definitive care



Secondary transfers avoided



Referral acceptance time



Data freshness (< 15 min)

Live demo: try it yourself in 3 minutes

- 1 Hospital staff signs in**
Bansal Hospital → BANSAL-ED01 → OTP shown on screen
- 2 Speak the emergency**
“Papa ko seene mein dard hai, pasina aa raha hai, 62 saal”
- 3 See what AI understood**
May indicate heart attack · CRITICAL; open “How was this decided?”
- 4 Find hospitals**
Open “Why this hospital?” and tap the match score
- 5 Prove the data is live**
Set ER to Diverting: the hospital vanishes from the phone instantly
- 6 Ask the hospital to accept**
Staff reads the SBAR note and accepts with bay Resus-02
- 7 Request the ambulance**
Watch it move on both live maps; preview the family link
- 8 Try “I’m alone”**
With a saved contact, all five steps run by themselves

Demo mode: hospital figures, SMS and ambulance dispatch are simulated and clearly labelled in the app.

Open it now

Live app

medreach-2.onrender.com

Report emergency

</report> (no login)

Citizen dashboard

</citizen>

Hospital dashboard

</hospital>

Source code

github.com/prakharpurwal518-collab/Medreach

Offline demo

<demo/medreach-prototype.html> (one file)

Why should Medreach be selected?



Real, urgent problem

Solves a life-or-death gap for citizens of Madhya Pradesh: the right hospital, the first time, with the bed confirmed before travel.



Working, deployed product

Both portals live on the web today, not slides or a mock-up. Judges can try the full flow in 3 minutes.



All 7 technologies

Generative AI, voice, vision, OCR, predictive and conversational AI in a mobile app, each in working code.



Safe & explainable AI

AI can only raise urgency; a tested engine decides and explains why. Never a diagnosis; 108 always one tap away.



Privacy by design

Consent, masked IDs, staff OTP and roles, audit logs and 24-hour deletion; aligned with the DPDP Act 2023.



Ready to scale

Low-cost open stack with a clear pilot plan and integration path to HFR, HMIS, 108 and ABDM.

Built for Madhya Pradesh first: Hindi voice, Ayushman / ABHA awareness, 108 always one tap away, and real Bhopal-region hospitals in the demo.

Conclusion: the right hospital, the first time

In an emergency, the question is not “where is the nearest hospital?” but “which hospital can treat this now, and has it said yes?”. Medreach answers it before the family leaves home.



The problem

Families go to the nearest hospital, find no specialist or bed, and lose the golden hour travelling again.



Our solution

AI understands the emergency; verified, time-stamped data finds a capable hospital; the hospital accepts before travel.



The proof

Deployed and working today: two portals, live maps, all 7 technologies, 51 automated tests, privacy by design.



Our ask: support a 3-month pilot with Bhopal hospitals, then integration with the Health Facility Registry, hospital HMIS, the 108 control room and ABDM for a state-wide rollout.

Live demo: medreach-2.onrender.com

Code: github.com/prakharpowal518-collab/Medreach Hospital login: Bansal Hospital · BANSAL-ED01

Thank you